Vulnerability Assessment Report

Target System: Metasploitable 2 Virtual Machine
Assessment Type: Network & Service Enumeration
Tools Used: Nmap, Kali Linux
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Executive Summary

A vulnerability assessment was conducted on the Metasploitable 2 virtual machine using Nmap from a Kali Linux environment. The objective of the assessment was to identify open ports, running services, operating system information, and potentially exploitable vulnerabilities.

The scan revealed multiple outdated and vulnerable services running on the target machine, including:

- vsftpd 2.3.4 FTP service
- UnrealIRCd IRC server
- Apache HTTP Server 2.2.8
- MySQL 5.0.51a
- Samba SMB services

Several of these services are publicly known to contain critical vulnerabilities that could allow attackers to gain unauthorized access, execute remote commands, or compromise the system.

The overall security posture of the target system is classified as: HIGH RISK

Immediate remediation and service hardening are recommended.

Assessment Objective

The purpose of this assessment was to:

- Identify open ports and active services
- Detect service versions
- Determine the operating system
- Identify known vulnerable services
- Evaluate overall security exposure

Scope

Item	Details
Target IP	10.0.2.4
Environment	Metasploitable 2 VM
Testing Machine	Kali Linux
Scan Type	Nmap Service & OS Detection
Command Used	`sudo nmap -sV -O 10.0.2.4`

Methodology

The following methodology was used during the assessment:

1. Reconnaissance: Basic information gathering on the target system.
2. Port Scanning: Identification of open TCP ports.
3. Service Enumeration: Detection of services and application versions using Nmap.
4. Operating System Detection: TCP/IP fingerprinting was used to identify the operating system.
5. Vulnerability Identification: Known vulnerable services were identified based on detected versions.

Tools Used

Tool	Purpose
Nmap	Port scanning & enumeration
Kali Linux	Penetration testing environment

Nmap Scan Results
Command Executed

sudo nmap -sV -O 10.0.2.4

Key Open Ports & Services

Port	Service	Version	Risk
21	FTP	vsftpd 2.3.4	HIGH
22	SSH	OpenSSH 4.7p1	Medium
23	Telnet	Linux telnetd	HIGH
25	SMTP	Postfix smtpd	Medium
80	HTTP	Apache 2.2.8	Medium
139/445	SMB	Samba 3.X–4.X	HIGH
3306	MySQL	MySQL 5.0.51a	HIGH
6667	IRC	UnrealIRCd	CRITICAL

Finding 1 — Vulnerable FTP Service

Vulnerability

vsftpd 2.3.4 Backdoor Vulnerability

Risk Level

🔴 HIGH

Description

The FTP service running on port 21 is using vsftpd version 2.3.4, a version known to contain a malicious backdoor vulnerability.
This vulnerability may allow attackers to gain unauthorized remote shell access to the system.

Business Impact

An attacker could:

- Gain remote access to the server
- Execute malicious commands
- Steal sensitive information
- Compromise the entire system

Remediation

- Disable outdated FTP services
- Upgrade to a secure FTP server version
- Use SFTP instead of FTP
- Restrict anonymous access

Finding 1

Evidence
Port 21 detected:

21/tcp open ftp vsftpd 2.3.4

Finding 2

Evidence
Port 6667 detected:

6667/tcp open irc UnrealIRCd

```
└$ sudo nmap -sV -O 10.0.2.4
[sudo] password for kali:
Starting Nmap 7.92 ( https://nmap.org ) at 2022-09-12 23:00 EDT
Nmap scan report for 10.0.2.4
Host is up (0.00045s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login       OpenBSD or Solaris rlogind
514/tcp  open  tcpwrapped
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:22:64:07 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
```

Finding 2 — UnrealIRCd Backdoor

Vulnerability

UnrealIRCd 3.2.8.1 Backdoor

Risk Level

🔴 CRITICAL

Description

The IRC service running on port 6667 uses UnrealIRCd, a version historically distributed with a malicious backdoor.

Attackers can exploit this vulnerability to execute arbitrary commands remotely.

Business Impact

Successful exploitation may allow:

- Remote code execution
- Full server compromise
- Malware installation
- Unauthorized administrative control

Remediation

- Immediately remove vulnerable IRC service
- Update to latest secure version
- Restrict unnecessary services
- Implement firewall filtering

Additional Security Weaknesses

Issue	Risk
Telnet Enabled	High
Old Apache Version	Medium
SMB Exposure	High
Outdated MySQL Version	High
Multiple Unnecessary Services	Medium

Security Concerns

The system exposes numerous outdated services that significantly increase the attack surface.

Legacy services such as Telnet transmit credentials in plaintext and should never be used in production environments.

Risk Summary

Severity	Count
Critical	1
High	4
Medium	3
Low	0

Overall Risk Rating

🔴 HIGH RISK

The target system contains multiple publicly known vulnerable services that may allow attackers to gain unauthorized access and compromise system integrity.

Recommendations

Immediate Actions

- ✓ Disable unnecessary services
- ✓ Remove Telnet service
- ✓ Upgrade vulnerable software
- ✓ Restrict open ports using firewall rules
- ✓ Apply latest security patches

Long-Term Improvements

- ✓ Perform regular vulnerability assessments
- ✓ Implement network segmentation
- ✓ Use secure protocols (SSH/SFTP)
- ✓ Monitor logs and suspicious activity
- ✓ Maintain asset inventory and patch management

Conclusion

The assessment identified several critical and high-risk vulnerabilities on the Metasploitable 2 system.
The presence of outdated and intentionally vulnerable services demonstrates how insecure configurations can expose systems to severe attacks, including remote code execution and unauthorized access.
Regular scanning, patch management, and secure service configuration are essential to maintaining a strong security posture.